

HELMET AND ALCOHOL DETECTION WITH IGNITION CONTROL

BY:

A. G. Kusuma – 23501A0406

G. Lalitha Vasavi – 23501A0437

G. Sujana sri – 23501A0439

G. Reshma sri – 23501A0442

K. Chaitanya sai – 23501A0463





INTRODUCTION: THE SAFETY CHALLENGE

This project responds to two leading contributors to motorcycle fatalities: riding without a helmet and driving under the influence of alcohol. The objective is to create an intelligent, fail-safe mechanism that prevents engine start when the rider is impaired or not wearing a helmet. The solution is non-punitive, immediate, and embedded at the vehicle level to provide proactive accident prevention.

- Primary goal: Block ignition if safety checks fail
- Design priority: Fast response, low false-trigger rate, and easy integration
- Validation: Software-in-the-loop and hardware simulation using Proteus and Keil

SOFTWARE USED

Keil uVision 4 (IDE & Toolchain)

- Used to write and compile Embedded C programs for the AT89C51 microcontroller.
- Provides an integrated environment for coding, compiling, debugging, and simulation.
- Detects and helps fix programming errors before hardware implementation.
- Generates a HEX file after successful compilation.
- The generated HEX file is loaded into Proteus to simulate the complete embedded system.
- Enables efficient development and testing of the Helmet Detection with Ignition Control and Alcohol Detection project.

Proteus 8 (Circuit Simulation)

- Used to design and simulate the complete embedded system circuit.
- Supports components such as the AT89C51, LCD, relay, LEDs, buzzer, and sensors.
- Allows loading the HEX file generated from Keil into the virtual microcontroller.
- Displays real-time circuit operation and signal behavior during simulation.
- Helps identify wiring, logic, and timing errors before hardware implementation.
- Reduces development time and improves overall system reliability.

CIRCUIT ARCHITECTURE

The proposed system is divided into three functional units: **Helmet Unit**, **Alcohol Unit**, and **Bike Unit**. These units work together to ensure that the vehicle operates only under safe riding conditions. By integrating embedded system technology, the project provides a reliable and efficient solution to enhance rider safety and reduce the risk of accidents.

Helmet Unit

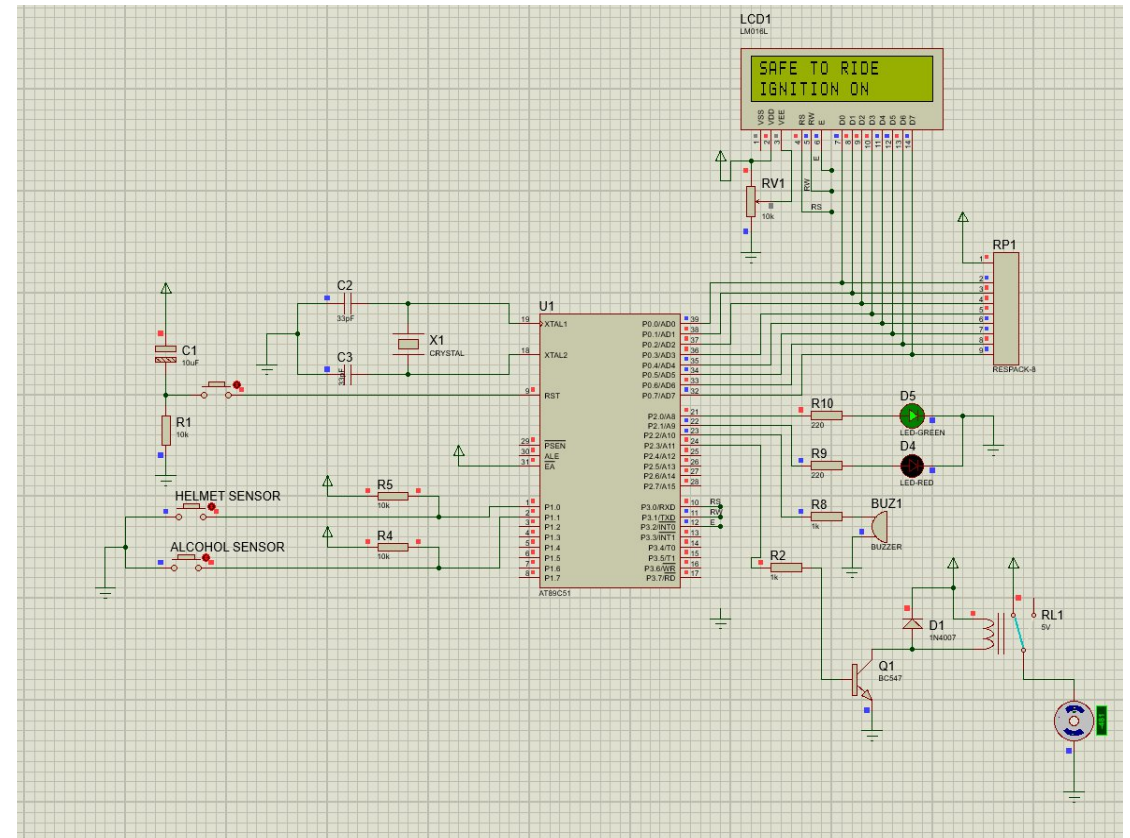
- Contains a helmet detection sensor, AT89C51 microcontroller, and RF transmitter.
- Detects whether the rider is wearing the helmet and sends the status to the bike unit.

Alcohol Unit

- Equipped with an MQ-3 alcohol sensor to detect alcohol in the rider's breath.
- Sends the alcohol detection status to the microcontroller for processing.

Bike Unit

- Consists of a AT89C51 microcontroller, relay, motor, LCD, LEDs, and buzzer.
- Processes helmet and alcohol status and enables ignition only if the helmet is worn and no alcohol is detected.



CIRCUIT COMPONENTS

- **AT89C51 Microcontroller** : The central processing unit that receives sensor inputs and controls the relay, LCD, LEDs, and buzzer.
- **Sensors (MQ-3 & Helmet Detection Sensor)** : The MQ-3 sensor detects alcohol in the rider's breath, while the helmet sensor verifies whether the helmet is worn.
- **16×2 LCD Display** : Displays system status and safety messages such as "Ignition ON", "Wear Helmet", and "Alcohol Detected".
- **Relay & BC547 Transistor** : The BC547 transistor drives the relay, which acts as an electronic switch to enable or disable the vehicle's ignition.
- **LEDs & Buzzer** : The **Green LED** indicates safe conditions, the **Red LED** indicates unsafe conditions, and the **buzzer** provides an audible warning.
- **Crystal Oscillator & Capacitors** : Generate a stable clock signal required for the proper operation of the AT89C51 microcontroller.
- **Resistors, Reset Switch & Power Supply** : Resistors protect the circuit and provide proper biasing, the reset switch restarts the microcontroller, and the regulated +5 V power supply powers the entire

KEIL uV4 EMBEDDED CODE

```
#include <REG51.H>

#define LCD P0

sbit RS = P3^0;
sbit RW = P3^1;
sbit EN = P3^2;

sbit HELMET = P1^0;
sbit ALCOHOL = P1^1;

sbit GREEN = P2^0;
sbit RED = P2^1;
sbit BUZZER = P2^2;
sbit RELAY = P2^3;

unsigned char previous_state = 255;

void delay(unsigned int ms)
{
    unsigned int i,j;
    for(i=0;i<ms;i++)
        for(j=0;j<1275;j++);
}

void lcd_cmd(unsigned char cmd)
{
    LCD = cmd;

    RS = 0;
    RW = 0;
    EN = 1;
    delay(2);
    EN = 0;

    if(cmd==0x01)
        delay(5);
}

void lcd_data(unsigned char dat)
{
    LCD = dat;

    RS = 0;
    RW = 0;
    EN = 1;
    delay(2);
    EN = 0;
}
```

```
void lcd_string(char *str)
{
    while(*str)
    {
        lcd_data(*str++);
    }
}

void lcd_init()
{
    lcd_cmd(0x38);
    lcd_cmd(0x0C);
    lcd_cmd(0x06);
    lcd_cmd(0x01);
    delay(10);
}

void displayHelmet()
{
    lcd_cmd(0x01);
    lcd_string("WEAR HELMET");
    lcd_cmd(0xC0);
    lcd_string("IGNITION OFF");
}

void displayAlcohol()
{
    lcd_cmd(0x01);
    lcd_string("ALCOHOL");
    lcd_cmd(0xC0);
    lcd_string("DETECTED");
}

void displaySafe()
{
    lcd_cmd(0x01);
    lcd_string("SAFE TO RIDE");
    lcd_cmd(0xC0);
    lcd_string("IGNITION ON");
}
```

```
void main()
{
    lcd_init();

    GREEN=0;
    RED=0;
    BUZZER=0;
    RELAY=0;

    delay(50);

    while(1)
    {
        delay(5);

        if(HELMET==0)
        {
            if(previous_state!=0)
            {
                GREEN=0;
                RED=1;
                BUZZER=1;
                RELAY=0;

                displayHelmet();

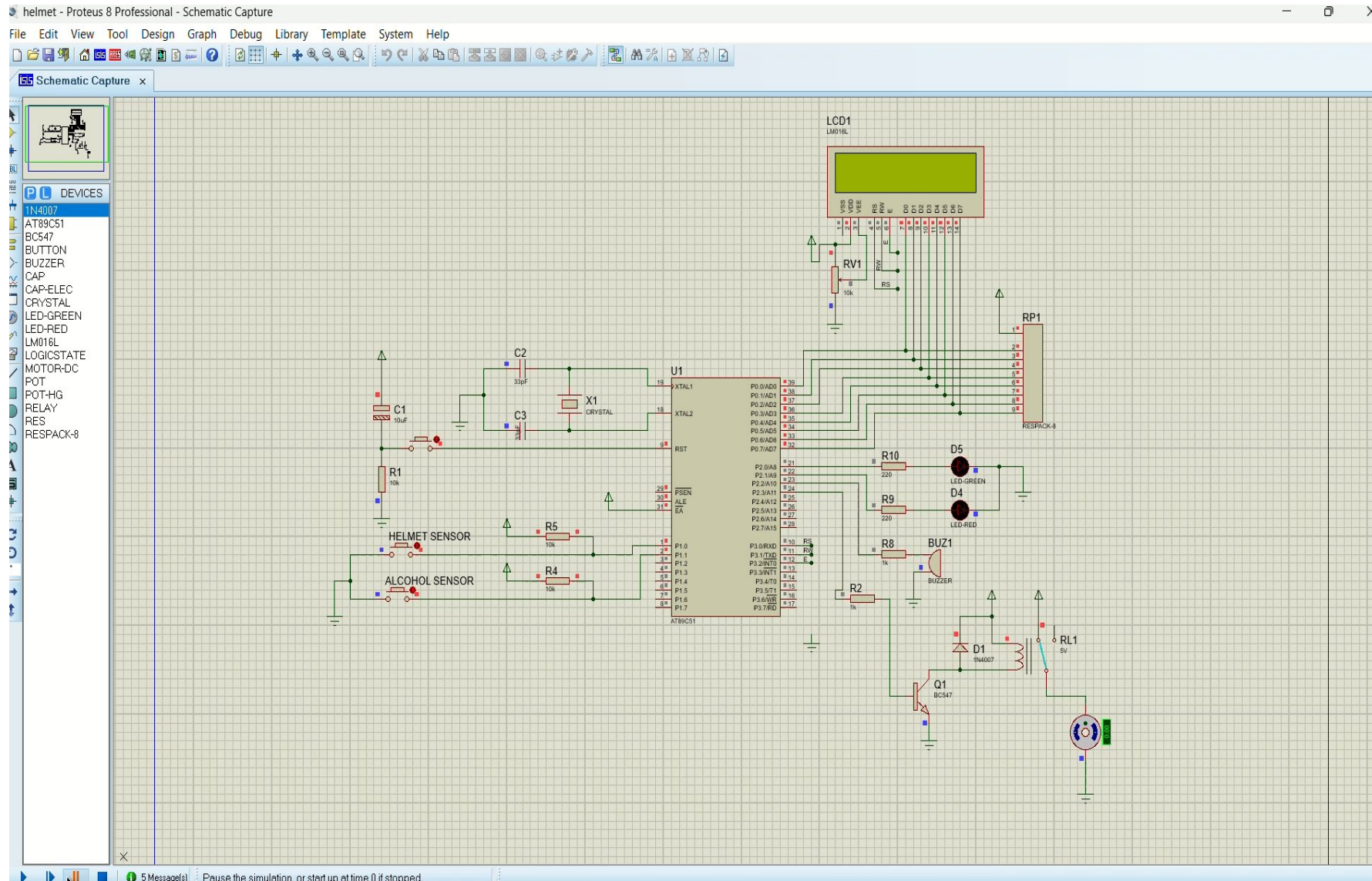
                previous_state=0;
            }
        }
        else if(ALCOHOL==0)
        {
            if(previous_state!=1)
            {
                GREEN=0;
                RED=1;
                BUZZER=1;
                RELAY=0;

                displayAlcohol();

                previous_state=1;
            }
        }
        else
        {
            if(previous_state!=2)
            {
                GREEN=1;
                RED=0;
                BUZZER=0;
                RELAY=1;

                displaySafe();

                previous_state=2;
            }
        }
        delay(10);
    }
}
```

PROTEUS
SIMULATION
&
WORKFLOW
RESULTS



USES

- Enhances rider safety by preventing unsafe vehicle operation.
- Reduces accidents caused by riding without a helmet or under the influence of alcohol.
- Suitable for motorcycles, scooters, and two-wheelers.
- Can be used for road safety awareness and smart transportation systems.

FEATURES

- Helmet detection before vehicle ignition.
- Alcohol detection using the MQ-3 sensor (or simulated input in Proteus).
- Automatic ignition ON/OFF control using a relay.
- Real-time status display on a 16×2 LCD.
- Audio-visual alerts using a buzzer and LEDs.
- Low-cost and reliable embedded system design.

EXTENSION OPPORTUNITIES

- Integrate GPS for real-time vehicle tracking.
- Add GSM to send emergency alerts to family members.
- Replace switches with actual helmet and MQ-3 sensors in hardware.
- Develop an IoT/mobile app for monitoring rider status.
- Add accident detection with automatic emergency notification.
- Implement fingerprint or RFID-based rider authentication



CONCLUSION

This project demonstrates how embedded systems can actively reduce road risk by combining sensor fusion, reliable decision logic, and ignition control. With rigorous calibration, simulation in Proteus 8, and disciplined code in Keil uV4, the design offers a practical, extendable platform to improve rider safety and accountability.



THANK
YOU